

Proper sub-Euler products violate Weil positivity on a fixed window: certified witnesses

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Abstract

Let $L = \log 11$ and let V be the 47-dimensional space of even test functions spanned by the shifted cosine basis on $[-L/2, L/2]$. We define, by explicit closed formulas, the pole, archimedean and prime-tower components of the truncated Weil quadratic form on V , and prove, by certified ball arithmetic (Arb), that *for every proper subset* $S \subsetneq \{2, 3, 5, 7\}$ of the primes interior to the window, the partial form Q_S (pole + archimedean – towers of S) admits an explicit negative witness: $Q_S(w_S) \leq -0.51$ in all fifteen cases, with certified enclosure radii below 10^{-8} . The complete form $Q_{\{2,3,5,7\}}$, by contrast, is numerically positive with smallest eigenvalue $\approx 3.6 \times 10^{-48}$ (not certified, and not claimed): removing any single prime tower flips the form from razor-edge positivity to $O(1)$ violation. Positivity of the truncated Weil form is thus, at this scale, a *quorum* of the full Euler product — no proper sub-product achieves it. The same engine certifies the quorum exhaustively at $\mu = 16$ (63/63 proper subsets violated) and $\mu = 22$ (255/255), and for the Dirichlet character χ_3 at $\mu = 11$ (7/7) — the phenomenon does not depend on the pole. In total: 340 certified violations, no exception. The result is unconditional and elementary in nature; it proves nothing about the Riemann hypothesis, but isolates, in certified form, the collective character of Weil positivity.

1 Setting and statement

Fix $\mu = 11$, $L = \log \mu$, and the interval $I_L = [-L/2, L/2]$. Let $\omega_k = 2\pi k/L$ and define the orthonormal *shifted cosine basis* of even-window test functions,

$$\eta_0 = \frac{1}{\sqrt{L}} \mathbf{1}_{I_L}, \quad \eta_k(u) = \sqrt{\frac{2}{L}} \cos(\omega_k(u + \frac{L}{2})) \mathbf{1}_{I_L}(u) \quad (1 \leq k \leq 46),$$

and $V = \text{span}(\eta_0, \dots, \eta_{46})$. For $f, g \in V$ and $x \geq 0$ set the symmetrized cross-correlation

$$\Theta_{f,g}(x) = \int f(u) g(u+x) du + \int f(u) g(u-x) du.$$

On basis pairs, $\Theta_{nm}(x) := \Theta_{\eta_n, \eta_m}(x)$ has the closed forms of Lemma 2 (Appendix A). Define on V the quadratic forms

$$\begin{aligned} P(f) &= \int_0^L \Theta_{f,f}(y) (e^{y/2} + e^{-y/2}) dy && \text{(pole),} \\ A(f) &= -\left[\frac{\Theta_{f,f}(0)}{2} C_L + \int_0^L \frac{e^{y/2} \Theta_{f,f}(y) - \Theta_{f,f}(0)}{e^y - e^{-y}} dy \right], \quad C_L = \gamma + \log \frac{4\pi(e^L - 1)}{e^L + 1} && \text{(archimedean),} \\ T_p(f) &= \sum_{k \geq 1, p^k \leq \mu} \frac{\log p}{p^{k/2}} \Theta_{f,f}(k \log p) && \text{(prime tower).} \end{aligned}$$

Since $\log 11 = L$ exactly and $\Theta_{f,f}(L) = 0$ identically (empty overlap), the tower T_{11} vanishes on V : the primes interior to the window are exactly $\{2, 3, 5, 7\}$. For $S \subseteq \{2, 3, 5, 7\}$ put

$$Q_S = P + A - \sum_{p \in S} T_p.$$

$Q_{\{2,3,5,7\}}$ is the truncated Weil form of ζ on the window (up to overall normalization; see the Remark below); the Q_S are its partial Euler sub-products.

Theorem 1 (Quorum at $\mu = 11$). *For every proper subset $S \subsetneq \{2, 3, 5, 7\}$ there is an explicit witness $w_S \in V$, given by 47 dyadic-rational coefficients (ancillary file), such that*

$$Q_S(w_S) < 0.$$

Certified upper bounds (ball arithmetic, enclosure radii $< 10^{-8}$):

S	bound	S	bound
$\emptyset$	−0.7745	$\{2, 3\}$	−0.6537
$\{2\}$	−0.7713	$\{2, 5\}$	−0.8420
$\{3\}$	−0.9076	$\{2, 7\}$	−0.9083
$\{5\}$	−0.9268	$\{3, 5\}$	−0.7822
$\{7\}$	−1.0514	$\{3, 7\}$	−1.0041
$\{2, 3, 5\}$	−0.5168	$\{5, 7\}$	−1.1524
$\{2, 3, 7\}$	−0.7101	$\{3, 5, 7\}$	−0.5421
$\{2, 5, 7\}$	−0.6804		

Remark. (i) The forms above are defined by the displayed formulas; the theorem is unconditional for these objects. Their identification with Weil’s pairing is exact and derived in Appendix B: for $f \in V$ with $g = f \star \tilde{f}$ and $\hat{g} = |\hat{f}|^2$, $Q_{\{2,3,5,7\}}(f) = \sum_{\gamma} |\hat{f}(\gamma)|^2$, the sum over the nontrivial zeros of ζ — with no normalization factor. In particular the Q_S are precisely the partial Euler sub-products of the Weil form. (ii) Nothing is claimed about $Q_{\{2,3,5,7\}}$: numerically its smallest eigenvalue is $\approx +3.58 \times 10^{-48}$, far below certification at working precision — appropriately so, since its positivity at all scales is equivalent to a window-by-window form of RH. The theorem’s content is the quorum: every proper sub-product fails by $O(1)$, while the complete product lands within 10^{-48} of zero on the positive side. (iii) Monotonicity fails: $\lambda_{\min}$ is not monotone in S (e.g. $\{5, 7\}$ is worse than $\emptyset$); a prime without its predecessors aggravates the violation.

Generalization

The certification engine is parametric in the scale and the L -function. Running it at $\mu = 16$ (interior primes $\{2, \dots, 13\}$, 63 proper subsets) and $\mu = 22$ ($\{2, \dots, 19\}$, 255 subsets) — same construction, with the cosine basis enlarged ($\dim V = 53$, resp. 57) to hold the conditioning, so these certified spaces are *not* nested subspaces of the 47-dimensional V of Theorem 1 —, and for the odd character χ_3 at $\mu = 11$ (no pole term; supported interior primes $\{2, 5, 7\}$, the archimedean part in its Dirichlet normalization) yields certified violation in *every* case: worst bounds -0.360 , -0.331 and -0.107 respectively, complete forms non-certifiable at $+0.000000$ throughout. Two regularities emerge, consistent with the recruitment structure of the quasi-radical: the mildest violation is always the deletion of the *last-recruited* prime, and its magnitude decreases with μ (-0.52 , -0.36 , -0.33); the most violent subsets consist of late primes deprived of their predecessors (e.g. $\{11, 13, 17, 19\}$ at $\mu = 22$: -1.86).

2 Certification

All matrix entries $P_{nm}, A_{nm}, (T_p)_{nm}$ ($0 \leq n \leq m \leq 46$) are enclosed in balls using Arb (via `python-flint` 0.9): tower entries are finite closed-form expressions in $\sin, \cos, \exp, \log$ evaluated in ball arithmetic; pole entries are rigorous integrals of entire integrands on $[0, L]$; archimedean entries are rigorous integrals on $[\varepsilon, L]$, $\varepsilon = 10^{-15}$, plus the tail bound of Lemma 1. The maximal enclosure radius over all 1128×7 entries is 9.4×10^{-11} . The witnesses w_S are the bottom eigenvectors of the double-precision midpoint matrices (their accuracy is immaterial: only the certified sign of the Rayleigh quotient matters); frozen as exact dyadic vectors, each quotient $Q_S(w_S)$ is a certified ball whose upper end is the bound in Theorem 1. Total computation: 7–16 seconds per scale with the parametric engine `quorum_general.py`, whose `verify` mode replays the frozen witnesses with no eigensolver dependence; the single-scale script `theoreme-quorum.py` reproduces the $\mu = 11$ table as a self-contained shortcut.

Lemma 1 (Tail bound). *For $0 \leq n \leq m \leq 46$ and $h_{nm}(y) = \frac{e^{y/2}\Theta_{nm}(y) - \Theta_{nm}(0)}{e^y - e^{-y}}$ one has, on $[0, 10^{-15}]$, $|h_{nm}(y)| \leq 250$, hence $|\int_0^{10^{-15}} h_{nm}| \leq 1000(n + m + 2) \cdot 10^{-15}$.*

Proof. From the closed forms, $|\Theta_{nm}| \leq 2$ and $|\Theta'_{nm}| \leq 2(\omega_n + \omega_m) + 4/L \leq 485$ on $[0, L]$. Hence $|\frac{d}{dy}(e^{y/2}\Theta_{nm})| \leq e^{y/2}(|\Theta'| + |\Theta|/2) \leq 487$ on $[0, 10^{-15}]$, so $|e^{y/2}\Theta_{nm}(y) - \Theta_{nm}(0)| \leq 487y$, while $e^y - e^{-y} \geq 2y$. Thus $|h_{nm}| \leq 244$, and the stated (deliberately lossy) bound follows. $\square$

3 Interpretation

The theorem gives the first certified instance of a phenomenon we call the *quorum*: truncated Weil positivity is a collective property of the complete Euler product within the window — each prime is individually indispensable, and partial products are not merely less positive but $O(1)$ -violating. Mechanically (lab notebook [7], file `report/le-milieu-des-premiers-v2.md`, §15), the violation is carried by the quasi-radical of the complete form: the tower of a removed prime p , restricted to that radical, has positive diagonal (the recruitment structure) but off-diagonal couplings of order 0.4 between rungs where p is diagonally silent and rungs where it is loud; a 2×2 minor then certifies a negative eigenvalue — *a prime cannot be silent on one rung while speaking to it from another without creating a negative direction*. The residual negatives of partial products are the radical-side face of the off-line pseudo-zeros of truncated Euler products. The complete form whose sub-products are certified here is the same object studied in the companion note [6]: its L^2 -normalized ground state converges to $\Phi/\|\Phi\|$, identifying the normalization constant of Suzuki's Conjecture 1.2 [5] as $\|\Phi\|_{L^2}$ — the quorum describes what holds that ground state up, one prime per rung. We conjecture the quorum at every scale μ and for every Dirichlet L -function; the general statement reduces (loc. cit.) to two harmonic-analysis assertions about the radical (diagonal silence before recruitment; survival of the coupling), neither of which mentions the Riemann hypothesis.

Reproducibility

Code, witnesses, certified-run script (7s) and the full lab notebook:

<https://github.com/darreal44/riemann-weil-phenomenology>

Verification record

(1) Appendix A verified by the author (and cross-checked at 10^{-30} against direct quadrature). (2) Appendix B derived in full for both the ζ and Dirichlet normalizations, with the public numerical cross-check `weil_normalization_check.py`. (3) The 340 witness vectors are frozen as exact dyadic rationals in `witnesses*.json`; the certification runs in `verify` mode from the frozen files, and the certificate tables carry the enclosure radii line by line. (4) Remaining: independent rerun of `quorum_general.py` `<mu>` `<NB>` 22 `<zeta|chi3>` `verify` on a second machine at publication time.

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A The correlation table

Lemma 2. *For $0 \leq n, m \leq 46$ and $0 \leq x \leq L$, with the shifted basis above,*

$$\Theta_{00}(x) = \frac{2(L-x)}{L}, \quad \Theta_{0m}(x) = -\frac{2\sin(\omega_m x)}{\sqrt{2}\pi m} \quad (m \geq 1),$$

$$\Theta_{nn}(x) = 2 \left[\frac{(L-x) \cos(\omega_n x)}{L} - \frac{\sin(\omega_n x)}{2\pi n} \right], \quad \Theta_{nm}(x) = \frac{2(n \sin(\omega_n x) - m \sin(\omega_m x))}{\pi(m^2 - n^2)} \quad (n \neq m, n, m \geq 1).$$

Proof. Substituting $v = u + L/2$ and $w = v - x$ in the two shift integrals gives $\Theta_{nm}(x) = \int_0^{L-x} [c_n(v)c_m(v+x) + c_n(v+x)c_m(v)] dv$ with $c_k(v) = \sqrt{2/L} \cos(\omega_k v)$ ($c_0 = 1/\sqrt{L}$); note $\omega_k(L-x) + \omega_k x = 2\pi k$, so all boundary sines evaluate through $\sin(2\pi k - \phi) = -\sin \phi$. *Case $n = m = 0$:* the integrand is $2/L$, giving $2(L-x)/L$. *Case $n = m \geq 1$:* product-to-sum, $\int_0^{L-x} \cos(2\omega_n v + \omega_n x) dv = -\sin(\omega_n x)/\omega_n$, whence $\Theta_{nn} = \frac{2}{L} [(L-x) \cos \omega_n x - \frac{L}{2\pi n} \sin \omega_n x]$. *Case $n = 0 < m$:* both integrals equal $-\sin(\omega_m x)/\omega_m$; the prefactor $\sqrt{2}/L$ gives $-2 \sin(\omega_m x)/(\sqrt{2} \pi m)$. *Case $1 \leq n \neq m$:* with $\Omega_{\mp} = \omega_n \mp \omega_m$, the two product-to-sum pairs contribute $2[\sin \omega_m x - \sin \omega_n x]/\Omega_- - 2[\sin \omega_n x + \sin \omega_m x]/\Omega_+$ (times $\frac{2}{L} \cdot \frac{1}{2}$); over the common denominator $\Omega_- \Omega_+ = \omega_n^2 - \omega_m^2$ the numerator collapses to $2\omega_m \sin \omega_m x - 2\omega_n \sin \omega_n x$, giving the stated formula. In the shifted basis no parity sign appears (the centered basis $\cos(\omega_k u)$ yields the same table multiplied by $(-1)^{n+m}$). All four formulas were verified against direct quadrature to 30 digits. $\square$

B Normalization: Q is exactly Weil's form

Lemma 3. *For $f \in V$, put $g = f \star \tilde{f}$ (so g is even, supported in $[-L, L]$, $\Theta_{f,f}(x) = 2g(x)$ for $x \geq 0$, and $\hat{g}(t) = |\hat{f}(t)|^2$ with $\hat{f}(t) = \int f(u) e^{itu} du$). Then*

$$P(f) = \hat{g}(\frac{i}{2}) + \hat{g}(-\frac{i}{2}), \quad \sum_{p \leq \mu} T_p(f) = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} g(\log n),$$

$$A(f) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{g}(t) \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) dt - g(0) \log \pi,$$

hence, by the Weil explicit formula, $Q_{\{2,3,5,7\}}(f) = \sum_{\gamma} |\hat{f}(\gamma)|^2$ over the nontrivial zeros $\frac{1}{2} + i\gamma$ of ζ .

Proof. The pole identity is immediate: $\int_0^L \Theta(y)(e^{y/2} + e^{-y/2}) dy = \int_{-L}^L g(y)(e^{y/2} + e^{-y/2}) dy = \hat{g}(i/2) + \hat{g}(-i/2)$ (verified to 12 digits). The prime identity is the definition of the towers. For the archimedean part, insert $\psi(z) = -\gamma + \sum_{k \geq 0} [\frac{1}{k+1} - \frac{1}{k+z}]$ at $z = \frac{1}{4} + \frac{it}{2}$ and use the Fourier pair $\frac{1}{2\pi} \int e^{itx} \operatorname{Re} \frac{1}{a+it/2} dt = e^{-2a|x|}$ ($a = k + \frac{1}{4}$): each term becomes $g(0)/(k+1) - \int g(x) e^{-(2k+\frac{1}{2})|x|} dx$. Writing $\frac{1}{k+1} = \int_0^\infty 2e^{-2(k+1)y} dy$ and summing the geometric series,

$$\frac{1}{2\pi} \int \hat{g} \operatorname{Re} \psi = -\gamma g(0) + \int_0^\infty \frac{2g(0)e^{-2y} - \Theta(y)e^{-y/2}}{1 - e^{-2y}} dy.$$

Two exact reductions bring this to the implemented form: replacing e^{-2y} by e^{-y} in the $g(0)$ -term (whose coefficient is $2g(0)$) costs $2g(0) \int_0^\infty \frac{e^{-y} - e^{-2y}}{1 - e^{-2y}} dy = 2g(0) \log 2 = g(0) \log 4$, and truncating at L (licit for Θ , which vanishes there) leaves the $g(0)$ -tail $g(0) \int_L^\infty \frac{dy}{\sinh y} = -g(0) \log \tanh(L/2)$. Collecting constants,

$$\frac{1}{2\pi} \int \hat{g} \operatorname{Re} \psi = -\frac{\Theta(0)}{2} [\gamma + \log(4 \tanh \frac{L}{2})] - \int_0^L \frac{e^{y/2} \Theta(y) - \Theta(0)}{e^y - e^{-y}} dy = A(f) + g(0) \log \pi,$$

since $C_L = \gamma + \log(4\pi \tanh \frac{L}{2})$ absorbs exactly the $-g(0) \log \pi$ of the explicit formula. (Numerical cross-check, public script `weil_normalization_check.py`: the pole identity holds to 12 digits; the archimedean identity to 3×10^{-6} at control height $T = 400$ and 2×10^{-7} at $T = 1600$, the residual being the control integral's own $O(1/T)$ tail.) $\square$

Lemma 4 (Dirichlet case). *For χ_3 (odd, conductor 3), with $s_0 = \frac{3}{4}$ and the completed L -function $(3/\pi)^{(s+1)/2} \Gamma(\frac{s+1}{2}) L(s, \chi_3)$, the implemented archimedean form*

$$A_\chi(f) = \frac{\Theta(0)}{2} C^{(3)} + \int_0^L e^{-2s_0 y} \frac{\Theta(0)e^{-(2-2s_0)y} - \Theta(y)}{1 - e^{-2y}} dy, \quad C^{(3)} = \log \frac{3}{\pi} - \gamma - \log(1 - e^{-2L}),$$

equals $\frac{1}{2\pi} \int \hat{g}(t) [\operatorname{Re} \psi(s_0 + \frac{it}{2}) + \log \frac{3}{\pi}] dt$ exactly.

Proof. As before, the ψ -series and the Fourier pair (now $a = k + s_0$) give

$$\frac{1}{2\pi} \int \hat{g} \operatorname{Re} \psi(s_0 + \frac{it}{2}) = -\gamma g(0) + \sum_{k \geq 0} \left[\frac{g(0)}{k+1} - \int g(x) e^{-2(k+s_0)|x|} dx \right] = -\gamma g(0) + \int_0^\infty \frac{\Theta(0)e^{-2y} - \Theta(y)e^{-2s_0y}}{1 - e^{-2y}} dy,$$

using $\frac{1}{k+1} = \int_0^\infty 2e^{-2(k+1)y} dy$ and two geometric sums. Here no $e^{-2y} \rightarrow e^{-y}$ substitution is made, so no $\log 2$ appears; truncating at L (licit: Θ vanishes there) leaves only the $\Theta(0)$ -tail $\Theta(0) \int_L^\infty \frac{e^{-2y}}{1 - e^{-2y}} dy = -\frac{\Theta(0)}{2} \log(1 - e^{-2L})$. Adding the conductor term $g(0) \log \frac{3}{\pi}$ reconstitutes $C^{(3)}$ and the integrand $e^{-2s_0y}(\Theta(0)e^{-(2-2s_0)y} - \Theta(y))/(1 - e^{-2y})$ exactly as implemented (note $e^{-2s_0y}e^{-(2-2s_0)y} = e^{-2y}$). The even case ($s_0 = \frac{1}{4}$) is identical. In particular the truncation constant $-\log(1 - e^{-2L})$, like $\log \tanh(L/2)$ in the ζ case, is not a convention but the exact tail of the window. $\square$

References

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